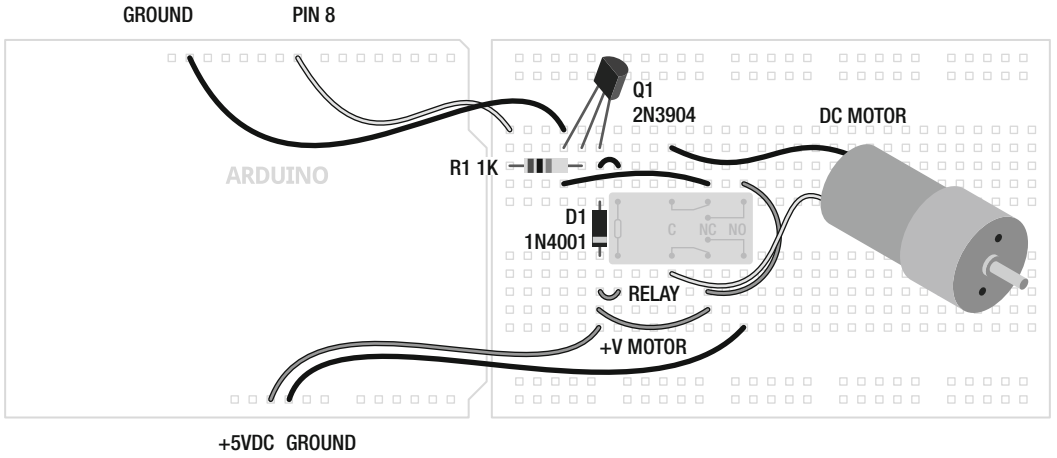

Figure 11-9. *Relay and DC Motor illustration*

This circuit is a little more complex, but it gives us bi-directional control of the motor using a single digital output, and we have a high amount of current to play with. Alas, this is at the cost of the ability to control the speed of the motor because the relay simply cannot switch at a very fast speed. Inside of the relay there are actually two switches with two positions each: normally open and normally closed. With the wiring as shown, when the relay is activated, the motor will be connected to the power supply in one direction and when the relay is off the motor will be connected in the opposite direction. This also means that the motor will always spin in one direction or another. This circuit doesn't quit!

While the coil of the relay is rated for +5v, I don't trust it with the microcontroller, so we've used a 2N3904 transistor to switch the relay on or off and a 1N4001 diode to protect the transistor from surges, like we did with the MOSFET. And finally, no two relays are ever really the same, so double-check the wiring for your relay before wiring this up.

That might have been a lot to take in, but the code is really simple and this brief little introduction to motors and the Arduino should be enough to send you on your way to find more information and give it a try.

Bonus Project 3: Mega-Size Something

Our final project idea isn't so much a project as it is a neat piece of hardware. Throughout this book, our projects have been written for the standard Arduino interface board, the Arduino Uno. The Uno, however, has a bigger brother called the Arduino Mega 2560, shown in Figure 11-10, and its capabilities should inspire new projects.